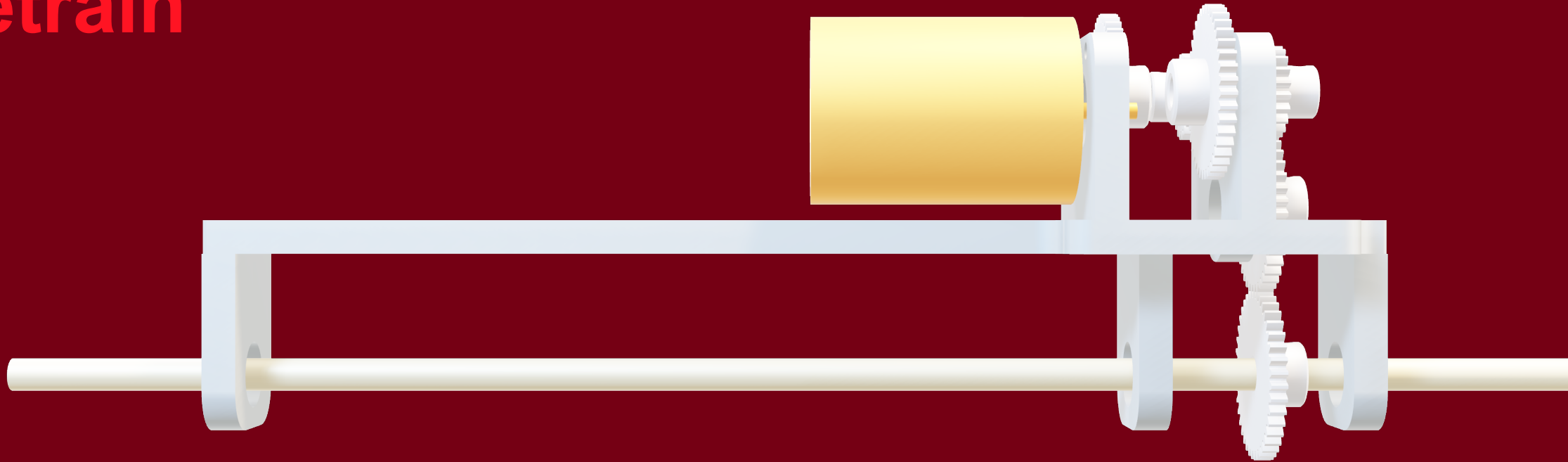




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Manufacturing Review

Drivetrain

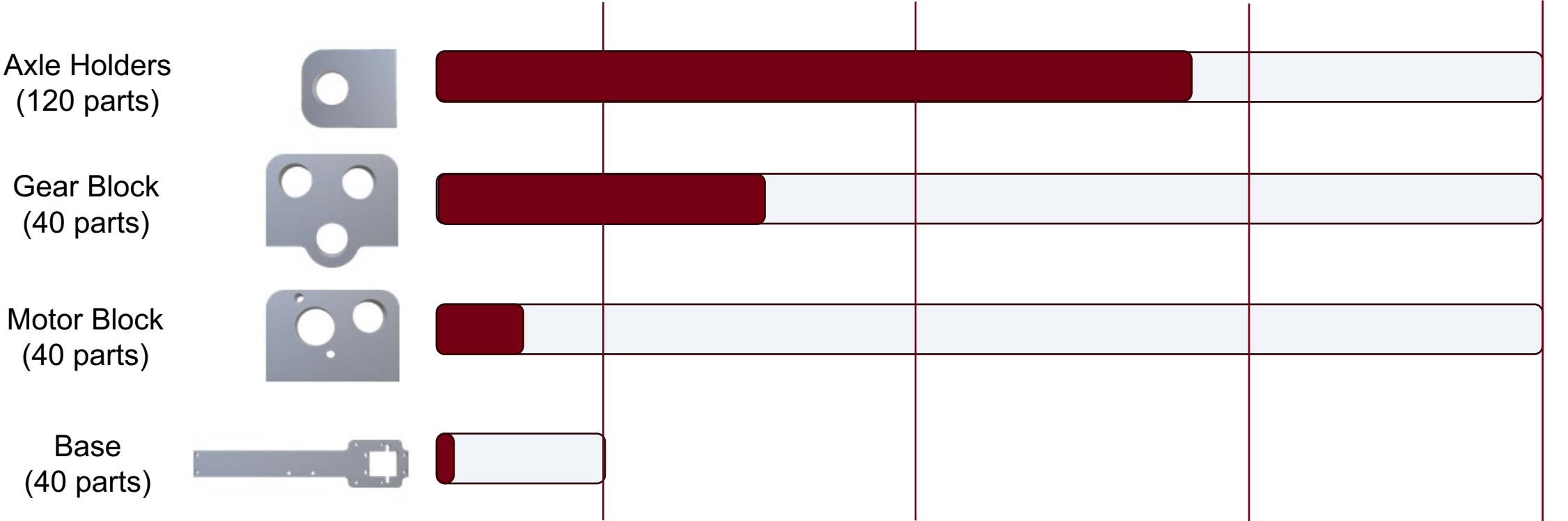


Mia Galatis, Sankaran Iyer, Sebastian Podiono,
Steve Rosario

November 12, 2025

Sub Assembly Team Overview

Current Status



- Extra Parts:
- Gears (265 parts) - Ream ✓
 - Small Axles (80 parts) - Cut
 - Medium Axles (40 parts) - Cut
 - Long Axles (40 parts) - Divot

Sub Assembly Team Overview

Challenge Addressed

Design Considerations

- Gear Ratio for Torque
- Base Mounting Locations
- Brushless Motor Wiring
- Gear Set Screw Loosening
- Axle Bending during Acceleration
- Gear and Bearing Tolerances

Design for Manufacturing

- Quality Control System for Bearing Press Fit
- Reduce Tapping areas
- Reduce Base size
- Minimize Tool Change – Line Design

Challenge Remaining

- Tap breaking when tapping

Design Considerations

Gear Ratio for Torque

Previous Design Considerations

Design Considerations

Gear Ratio Analysis & Design Decisions

Objective: navigating

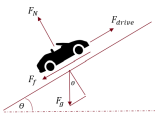
Motor Considerations:

- Motor: F
- Input Vc
- Motor R
- Motor sh
- pinion ge

Design Considerations

Ramp Climbing Feasibility Analysis

Objective: Determine whether the motor provides sufficient torque for the vehicle to climb the ramp based on certain assumptions.



$$F_{drive} = mg \sin \theta + \mu mg \cos \theta$$
$$T_{motor} = \frac{P_{in}}{\omega} = \frac{21.6W}{(26640 \text{ RPM} \cdot \frac{2\pi}{60})} = 0.00774 \text{ Nm}$$
$$T_{axle, req} = \frac{F_{drive} r_{wheel}}{\text{Gear Reduction}}$$
$$T_{axle, avail} = T_{motor} \cdot \text{Gear Reduction}$$

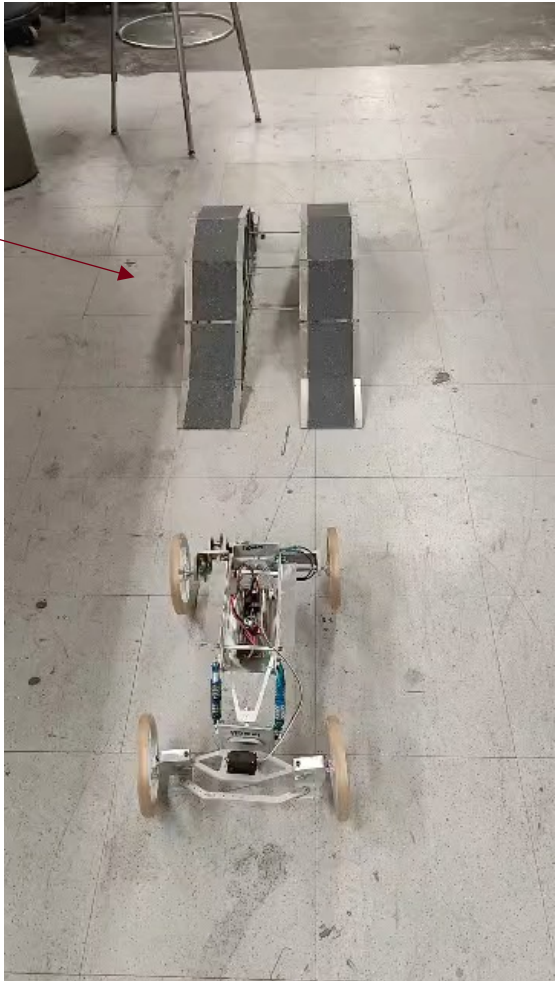
Mass			
3 lbs. (~1.36 kg)	10.35	0.059	0.103
5 lbs. (~2.27 kg)	17.28	0.099	0.103

Assumptions:

- Wheel diameter (D = 6")
- Gravity (g = 9.81 m/s²)
- RC car mass cases: 3 lbs., 5 lbs.
- Fraction coefficient: 0.1
- Maximum Ramp Angle: 45°
- Max Battery Power: 21.6 W

Massachusetts Institute of Technology

Testing indicates that the RC car can sufficiently pass through the ramp



Calculation from past presentation location in the appendix

Mass	F_{gx} (N)	$T_{axle, req}$ (Nm)	$T_{axle, avail}$ (Nm)
3 lbs. (~1.36 kg)	9.43	0.049	0.103
5 lbs. (~2.27 kg)	15.75	0.083	0.103
6.8 lbs. (~3.08 kg)	21.37	0.112	0.103

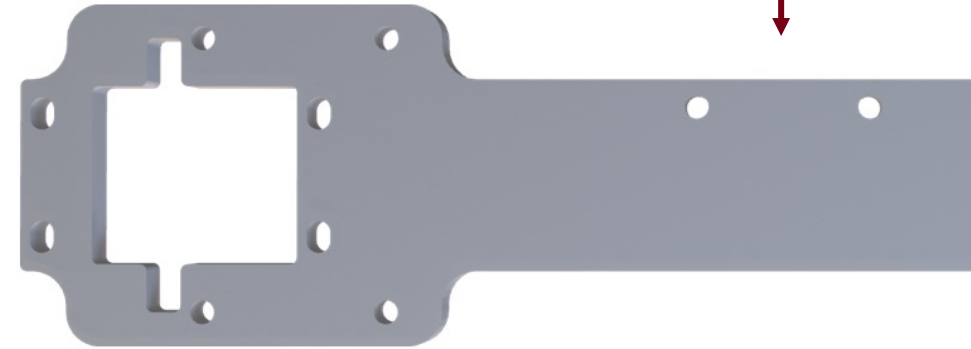
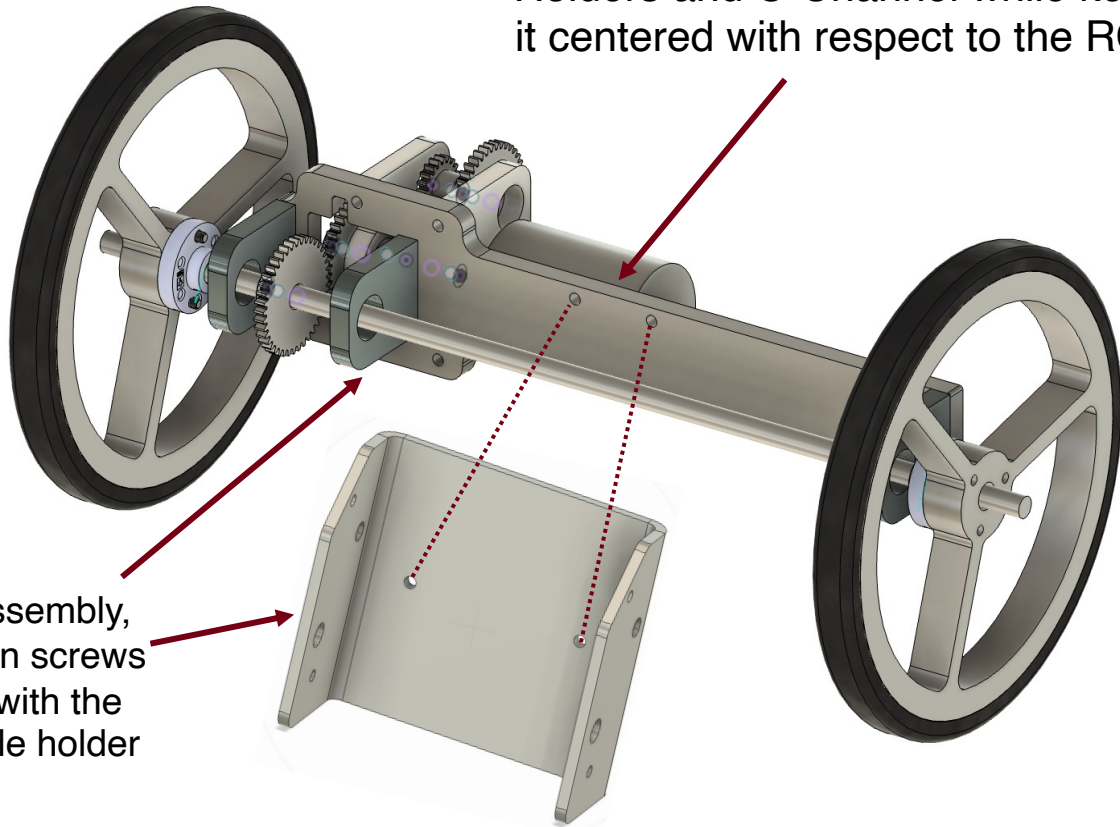
Weight of Prototype

Design Considerations

Base Mounting Locations

Base plate was slightly widened
prevent collision between the Axle
Holders and C-Channel while keeping
it centered with respect to the RC Car

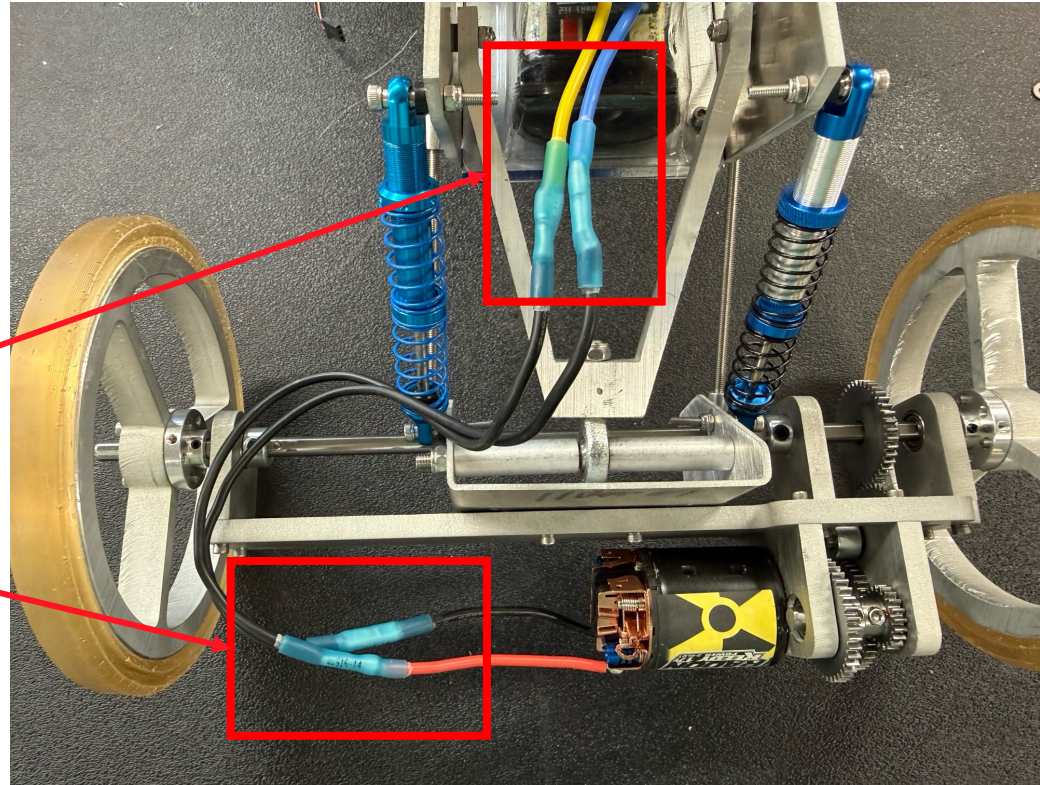
During assembly,
suspension screws
clashed with the
middle axle holder



Design Considerations

Brushless Motor Wiring

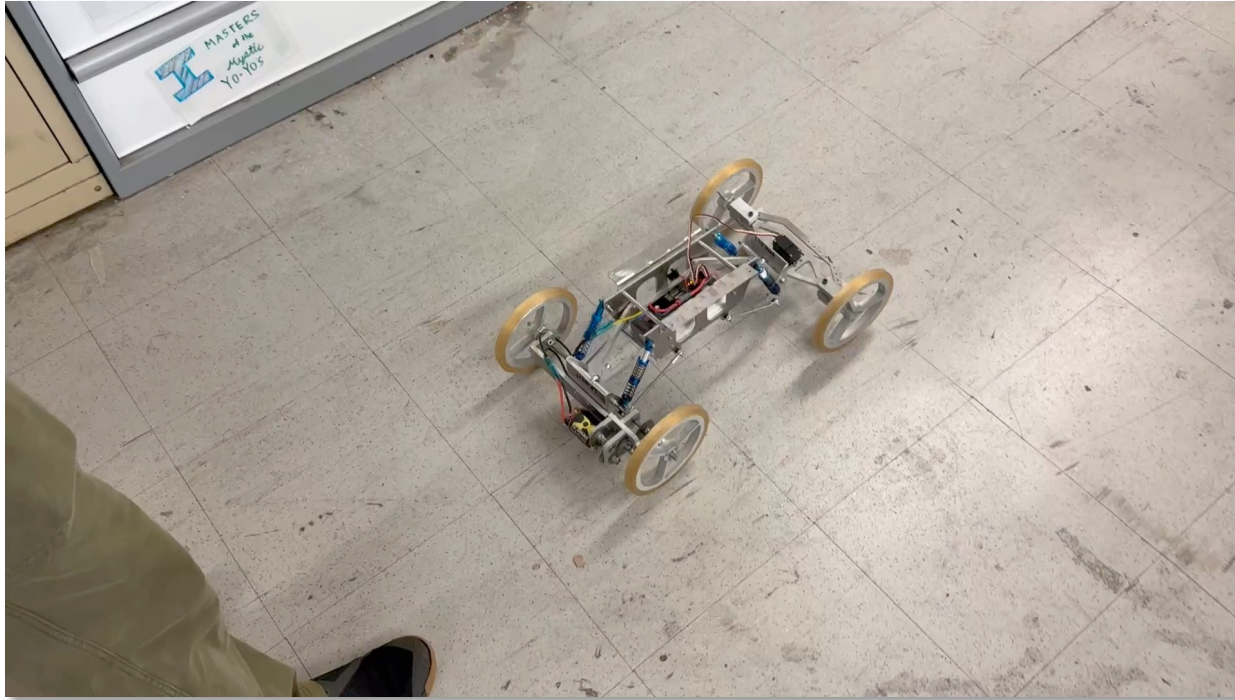
Existing motor & battery wire length was insufficient for routing, so extension wires were connected and crimped in.



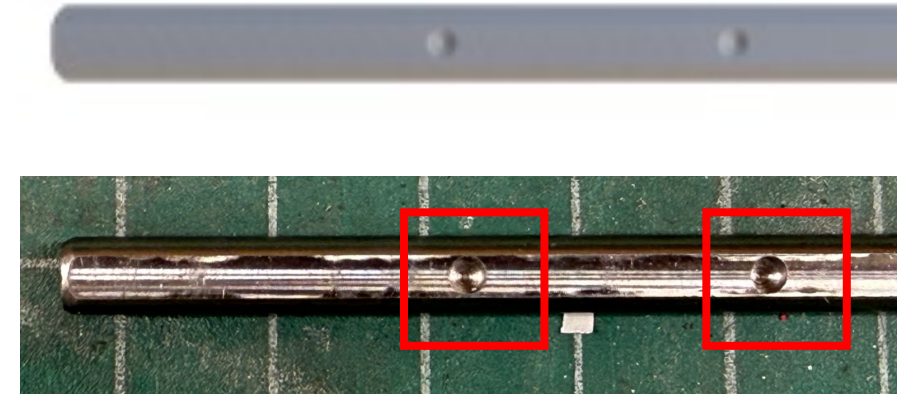
Wires & Connectors
ordered for final assembly
of 40 RC Cars

Design Considerations

Gear Set Screw Loosening



Prototype testing indicated that vibrations and collisions caused set screws holding the wheels and gears to come loose and slip.



- Drivetrain gears secured using purple thread locker (Loctite)
 - Weakest available thread locker used to ensure set screws are removable using hand tools to keep gears reusable
 - 24-hour cure time for thread locker
- Divots drilled into main axle to secure drive gear and rear wheels

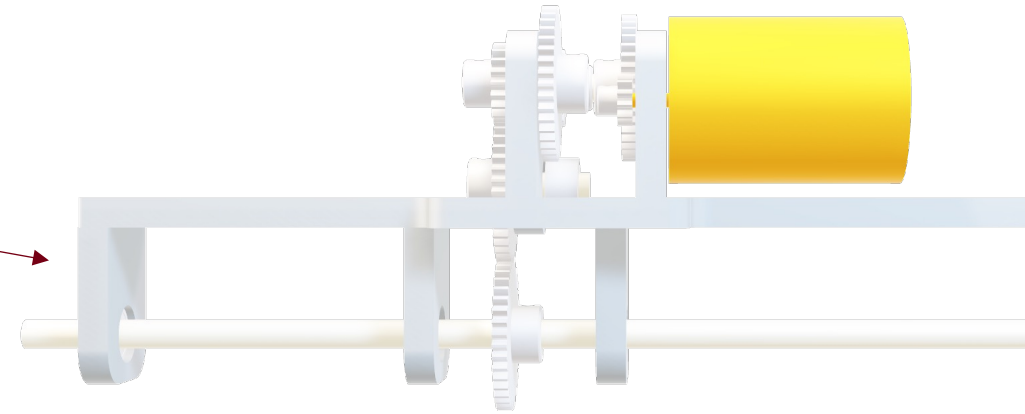
Design Considerations

Axle Bending

Design Considerations:

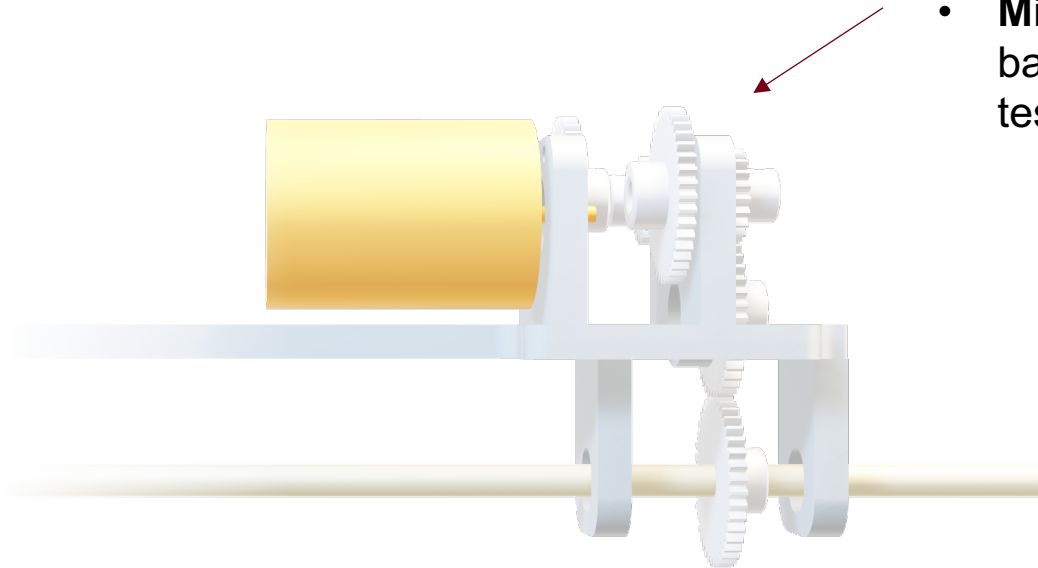
Axle holders designed to be identical to prevent rod deflections and allow for simple manufacture

Initial design housing 4 axle holders

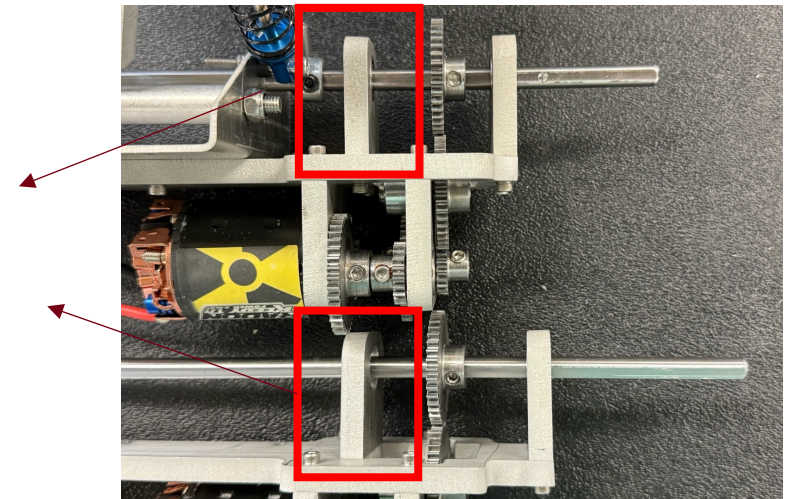


Updated Design based on Final Drive Gear Position:

- **Removed redundant axle holder** based on shift in drive gear position
- **Middle axle holder position was optimized** based on axle bending observations during testing



Axle Holder moved closer to avoid deflection



Design Considerations

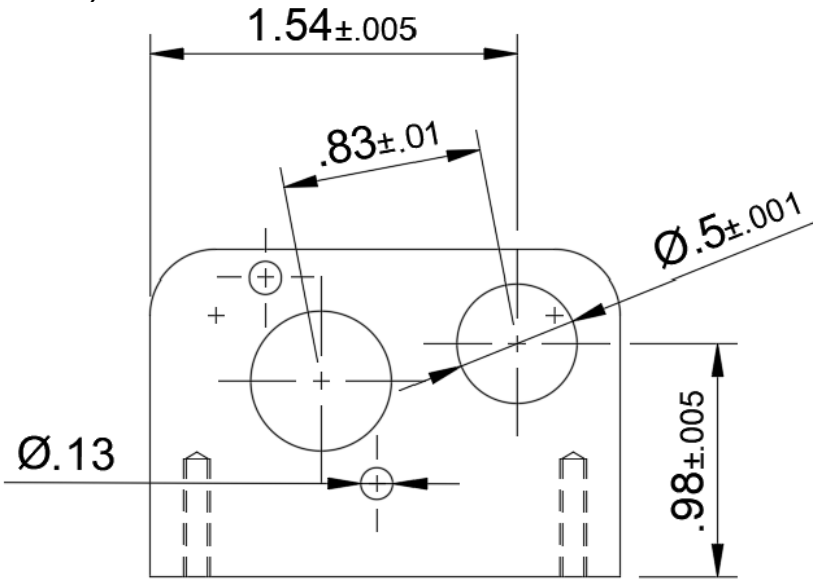
Gear and Bearing Precision Tolerance



OMAX 2652 WATERJET

- LINEAR POSITIONAL ACCURACY: $\pm 0.001''$ (± 0.025 mm)
- REPEATABILITY: $\pm 0.001''$ (± 0.025 mm)
- BALLBAR CIRCULARITY: $\pm 0.003''$ (± 0.076 mm)

Press Fit Type	Application	Interference (inch)
Light	Small bearings, easy assembly/removal	0.0002"–0.0004"
Medium	Normal load, general use	0.0004"–0.0006"
Heavy	Shock/vibration, high load	0.0006"–0.0010"

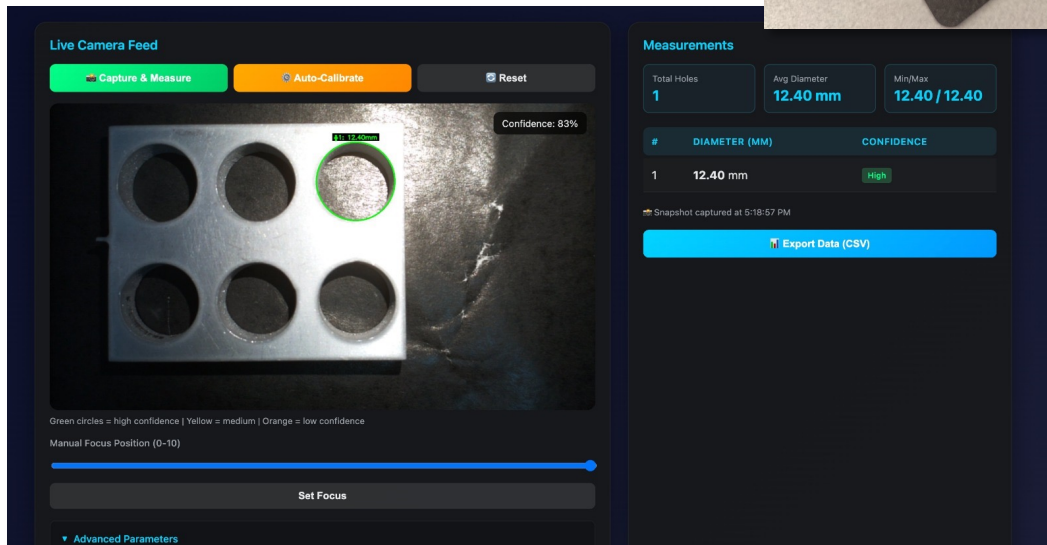
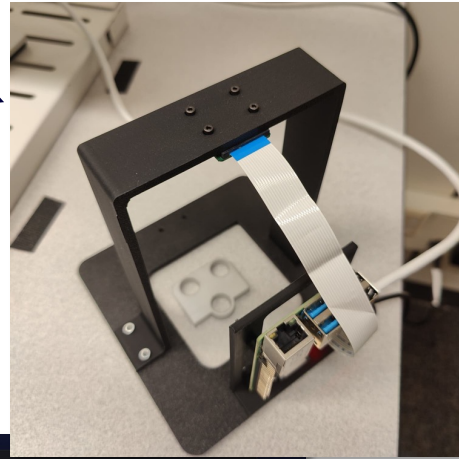


Design for Manufacturing

Quality Control System for Bearing Press Fit

Raspberry Pi
Camera Module
Set-Up

Web App
Interface



Vision System Setup

- Used Raspberry Pi camera + OpenCV for hole measurement
- Parts placed under camera, multiple measurements taken to estimate hole diameter
- **Data automatically logged** to process measurement file and **run charts**

Current State

- Basic OpenCV measurement pipeline implemented
- System detects and measures target features but is influenced by lighting and positioning parameters

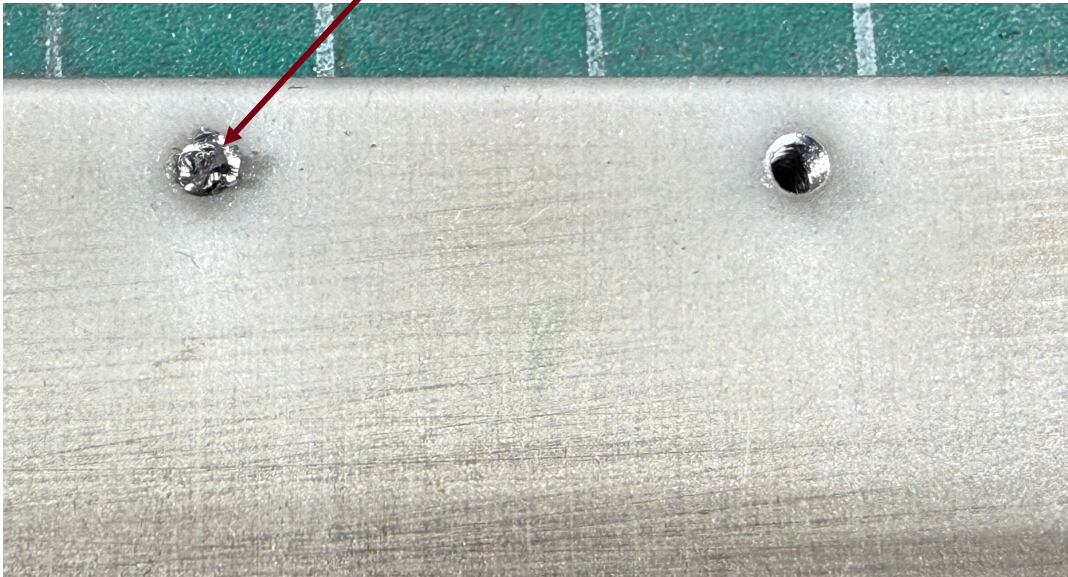
Challenges & Learnings

- Emphasized need for data validation and iterative tuning
- Continued improvements required for repeatability and accuracy

Design for Manufacturing

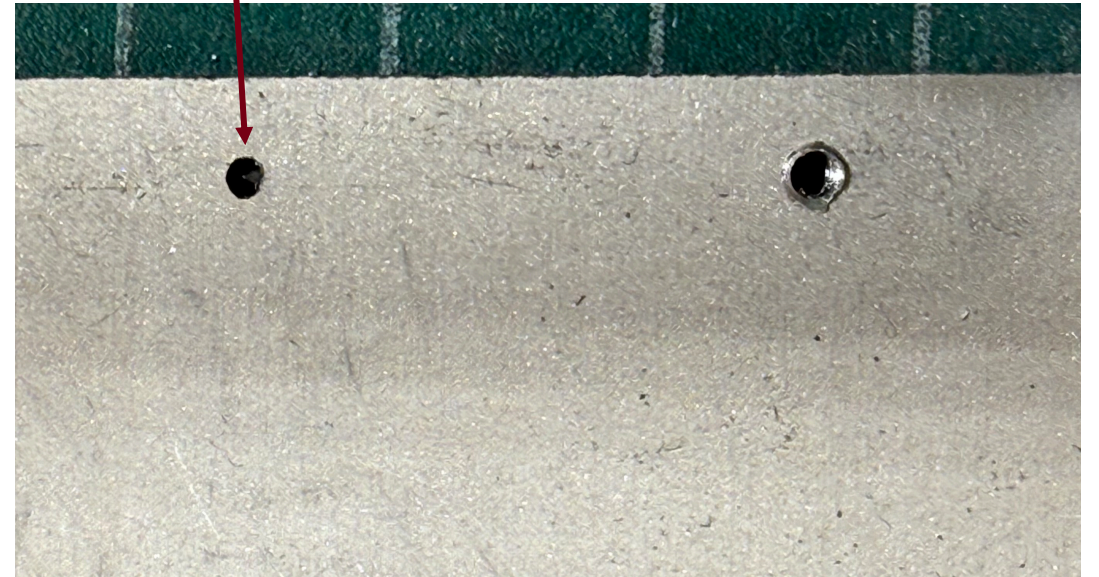
Reduce Tapping Areas

Waterjet mounting holes were often too small to be tapped, resulting in several broken taps



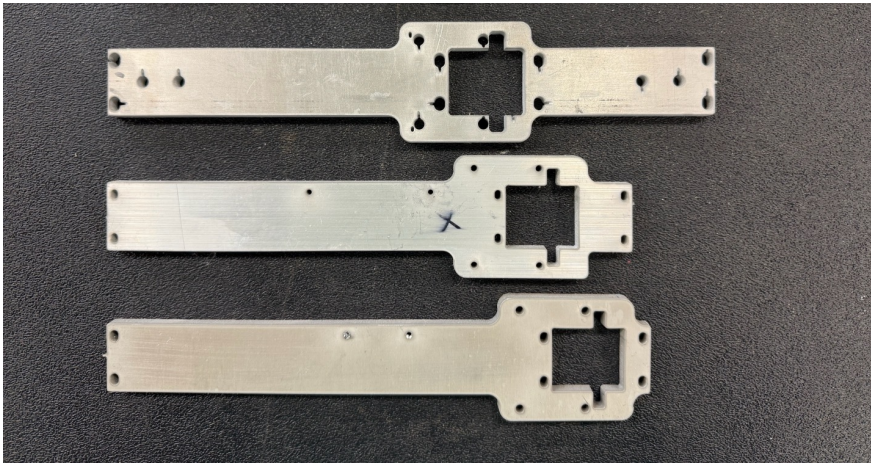
Tapped holes were changed to clearance holes

- Suspension and drivetrain now mated using 4-40 screws and 4-40 locknuts.
- Prevents taps from breaking
- Avoids an extra drilling/reaming step for the base



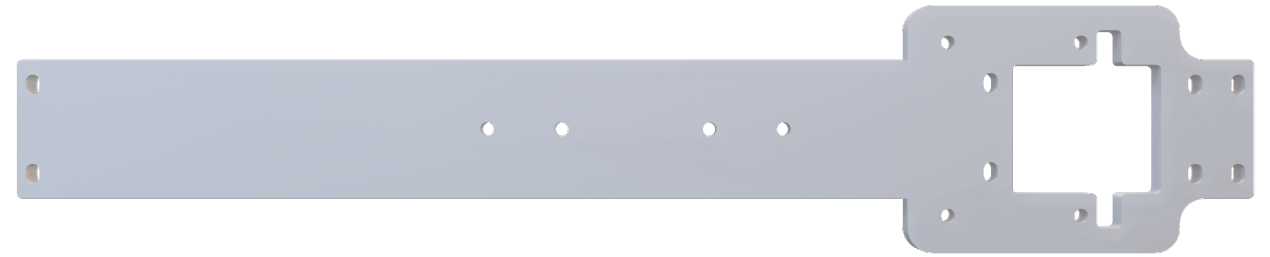
Design for Manufacturing

Reduce Base Size

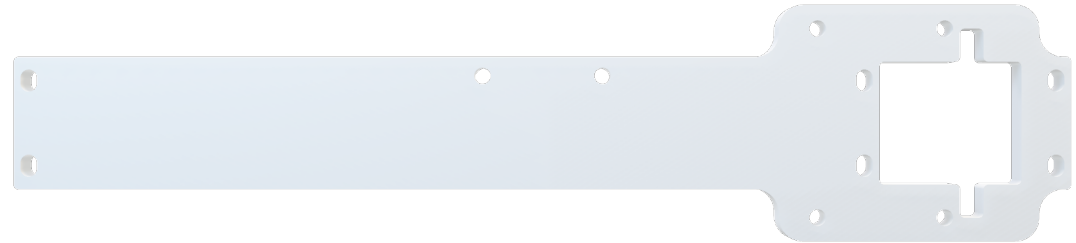


Base length shortened in each design iteration

- Helps steering maintain a tight turning radius
- Reduces waterjet cutting time (7 min to 4 min)



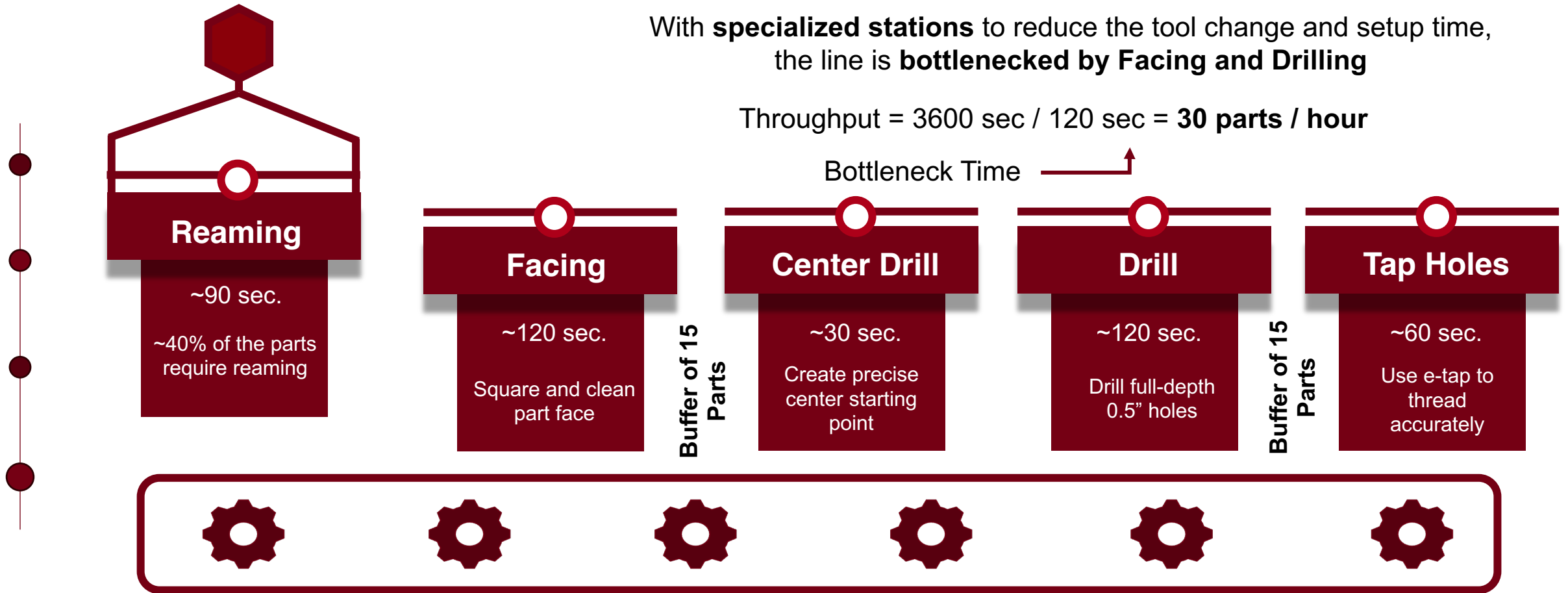
250 mm (9.84")



233 mm (8.8")

Design for Manufacturing

Minimize Tool Change – Line Design



Note: We will be using a vise stop to set our zero and avoid using edge finder for each part

Cycle Time Analysis – Unoptimized Line

#	Operation	Machine	Tooling	Cycle time (min:sec)
1	Cut stock to shape	Water jet	Water jet	0:29
2	Check the quality of the bearing hole	RasPi Camera	RasPi Camera	0:02
3	Fixture part and edge find	Mill	Vise, edge finder	1:12
4	Tool change from edge finder to reamer			0:22
5	Ream Bearing Holes If < 0.498"	Mill	0.498" reamer	1:42
	Tool change from edge finder to reamer			0:21
	Refixture part and edge find	Mill	Vise, edge finder	1:15
6	Tool change from edge finder to endmill			0:15
7	Face the bottom of the workpiece	Mill	½" endmill	1:56
8	Tool change from end mill to center drill			0:08
9	Center drill holes	Mill	#2 60 degrees Center drill	0:38
10	Tool change from center drill to drill			0:11
11	Drill holes	Mill	#43 drill	2:26
12	Tap holes	Powered tap	4-40 tap	1:03

Cycle Time Analysis – Optimized Line

#	Operation	Machine	Tooling	Cycle time (min:sec)
1	Cut stock to shape	Water jet	Water jet	0:29
2	Check the quality of the bearing hole	RasPi Camera	RasPi Camera	0:02
3	Ream Bearing Holes If < 0.498"	Mill 1	0.498 in. reamer	1:42
4	Face the bottom of the workpiece	Mill 2	½" endmill	1:56
5	Center drill holes	Mill 3	#2 60 degrees Center drill	0:38
6	Drill holes	Mill 4	#43 drill	2:26
7	Tap holes	Powered tap	4-40 tap	1:03

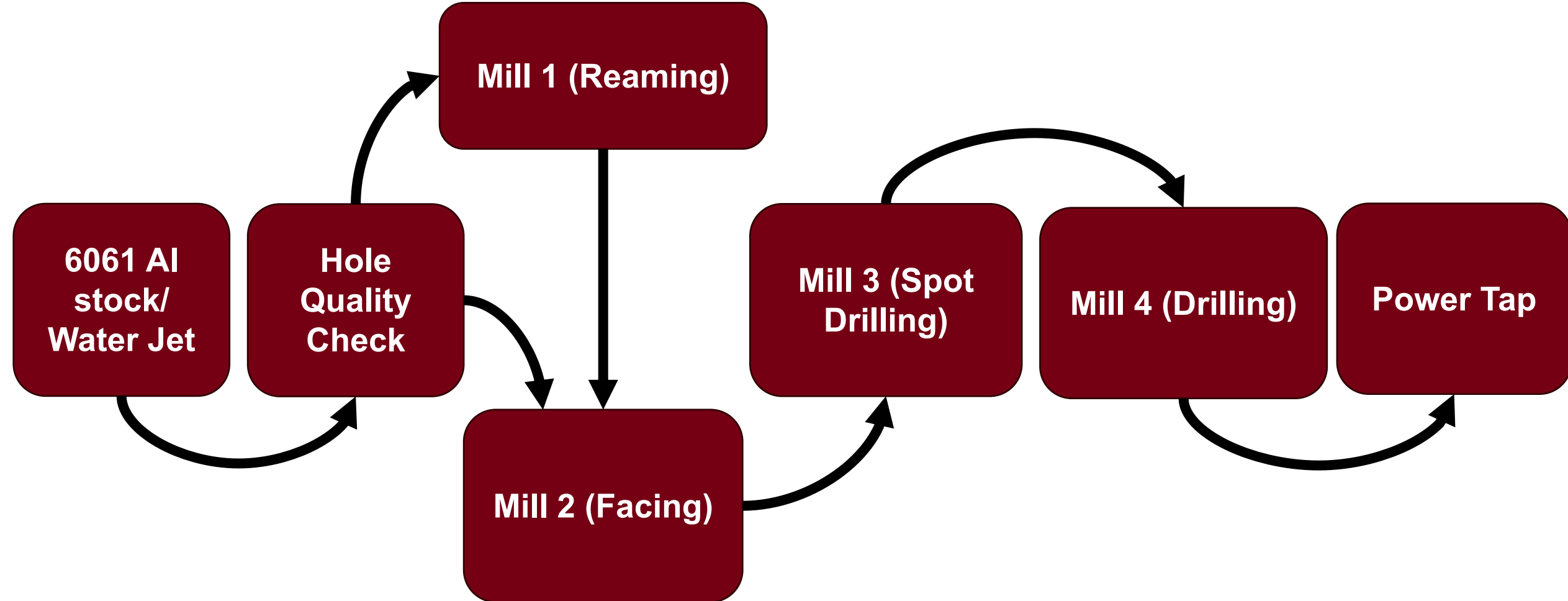
Unoptimized Line = Vehicles * (Value adding operations + Fixturing time + Tool Change time)
 = 120 * (451 sec + 147 sec + 77 sec) = 81000 sec = 22.5 hours

Optimized Line = Vehicles * (Value adding operations) + Fixturing time
 = 120 * (451 sec) + 297 = 54417 sec = 15.1 hours

7.4 hours saved for 120 Axle Holders (for 40 RC Cars)!

High Level Process Plan

Axle Holder, Gear Block, Motor Mount



High Level Process Plan

Base



Gears



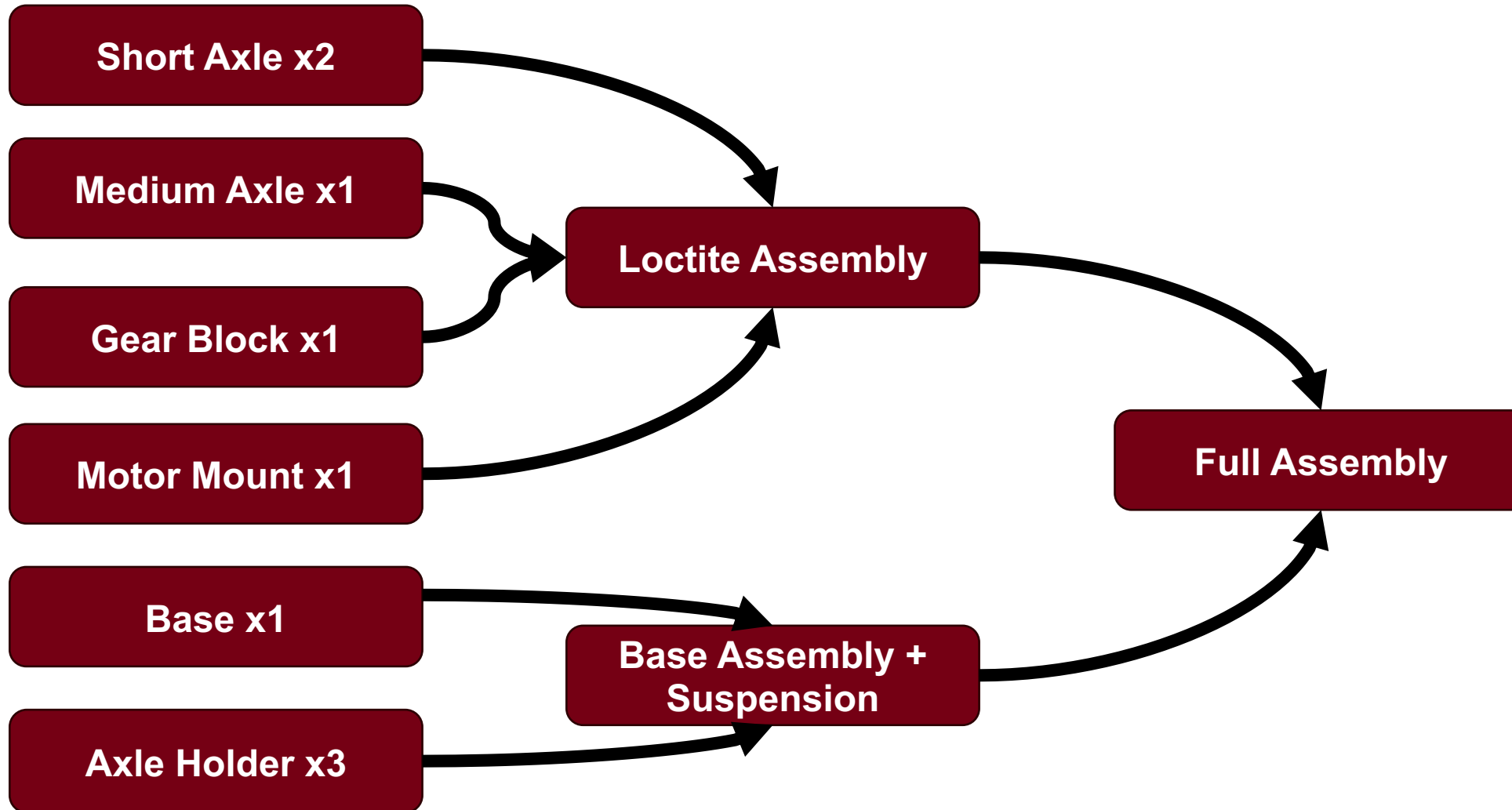
Short + Medium Axles



Long Axles

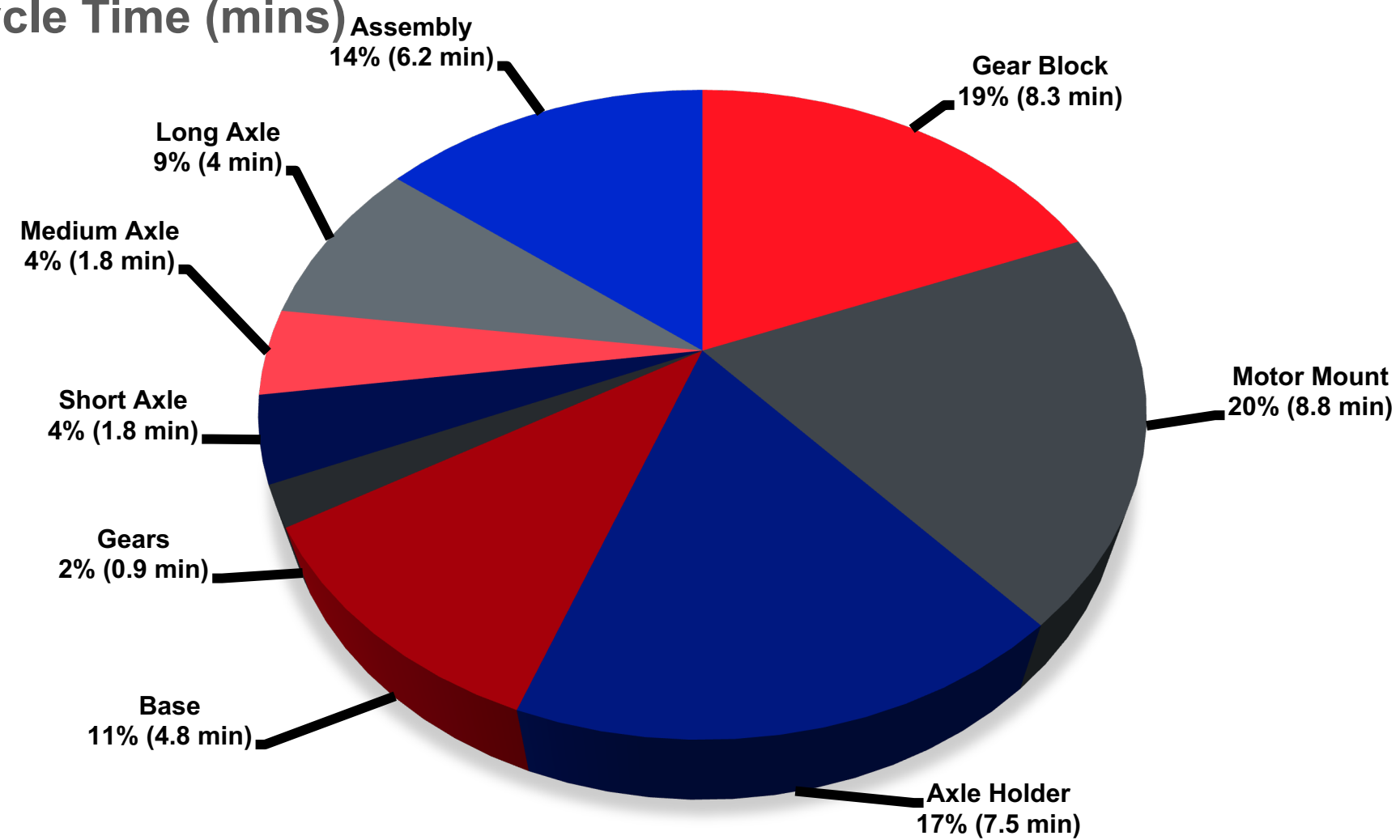


High Level Assembly Process Plan



Analysis of Manufacturing Process

Cycle Time (mins)



Total cycle time for 1 unit = **44 min**

* Cycle times for each step detailed in Appendix A

Cost per Unit

Part	Material	Cost
Base	6061 Aluminum	\$5.06
Axle Holder	6061 Aluminum	\$1.13
Gear Block	6061 Aluminum	\$1.18
Motor Mount	6061 Aluminum	\$0.64
Bearing	Chrome Steel	\$6.39
Shaft Collar	Carbon Steel	\$5.49
10-32 Set Screw	18-8 Stainless Steel	\$0.67
4-40 x 1/8 Set Screw	18-8 Stainless Steel	\$0.09
4-40 x 1/2 Screw	18-8 Stainless Steel	\$0.86
4-40 Cone Point Set Screw	18-8 Stainless Steel	\$0.25
Bullet connectors	Brass	\$0.36
16 AWG Wire	Copper	\$0.17
Gears, axles, and motor provided by 2.810		

Total cost for 1 unit = \$22.31

*** Detailed BOM located
in Appendix B**

Projections for 40 units

Total time = 40 units * 44 min/unit = 1760 min =
29.33 hours

Total cost = 40 units * \$22.31 min/unit =
\$892.40

Cost and Time Projections

Projections for 1,000,000 units

Item	Basis	Estimated Unit Cost (after scale efficiencies)	Total Cost (1,000,000 units)
Raw Materials	Aluminum	\$8.85 /unit ($\approx$ 15% bulk discount)	\$8.85 M
Components	Bearings, motors, connectors, screws, etc.	\$42.70 /unit ($\approx$ 10% discount)	\$42.7 M
Packaging + Logistics	Boxes, labels, freight	\$0.60 /unit	\$0.6 M
Labor (Assembly + QA)	7 min /unit at \$30/hr	\$3.50 /unit	\$3.5 M
Overhead	5 % of total direct cost	-	\$1.1 M

Total Projected Cost: $\approx$ \$57.75M

Per-unit cost $\approx$ \$55.65 / unit (vs. \$22.31)

Expected margin: $\sim$ 5-10% cost increase due to industrial overhead and potential changes in manufacturing processes, but 15% supplier savings (bulk ordering of material)

Time Projection: Full-scale rate: 4,000 units/day $\rightarrow$ 1,000,000 units in $\sim$ 250 days ($\approx$ 12 months).
With two lines: Production can be completed in $\sim$ 6 months.

Inventory

Item	Price	Qty Purchased	Qty Available
Bearing	\$271.66	360	312
Shaft Collar	\$196.80	120	102
10-32 Flat Set Screw	\$33.52	400	346
4-40 x 1/8 Set Screw	\$9.36	40	34
4-40 x 1/2 Screw	\$36.96	600	516
4-40 Taps	\$43.05	5	3
4-40 Lock Nuts	\$3.85	100	100
10-32 Cone Point Set Screw	\$50.60	200	199
4-40 Chip Clearing Taps	\$66.56	2	2
Purple Thread Locker	\$19.98	2	2
16 AWG Wires	\$17.39	1	1
Bullet Connectors	\$8.99	400	396

Budget Remaining = \$1,000.00 – \$758.72 = \$241.28

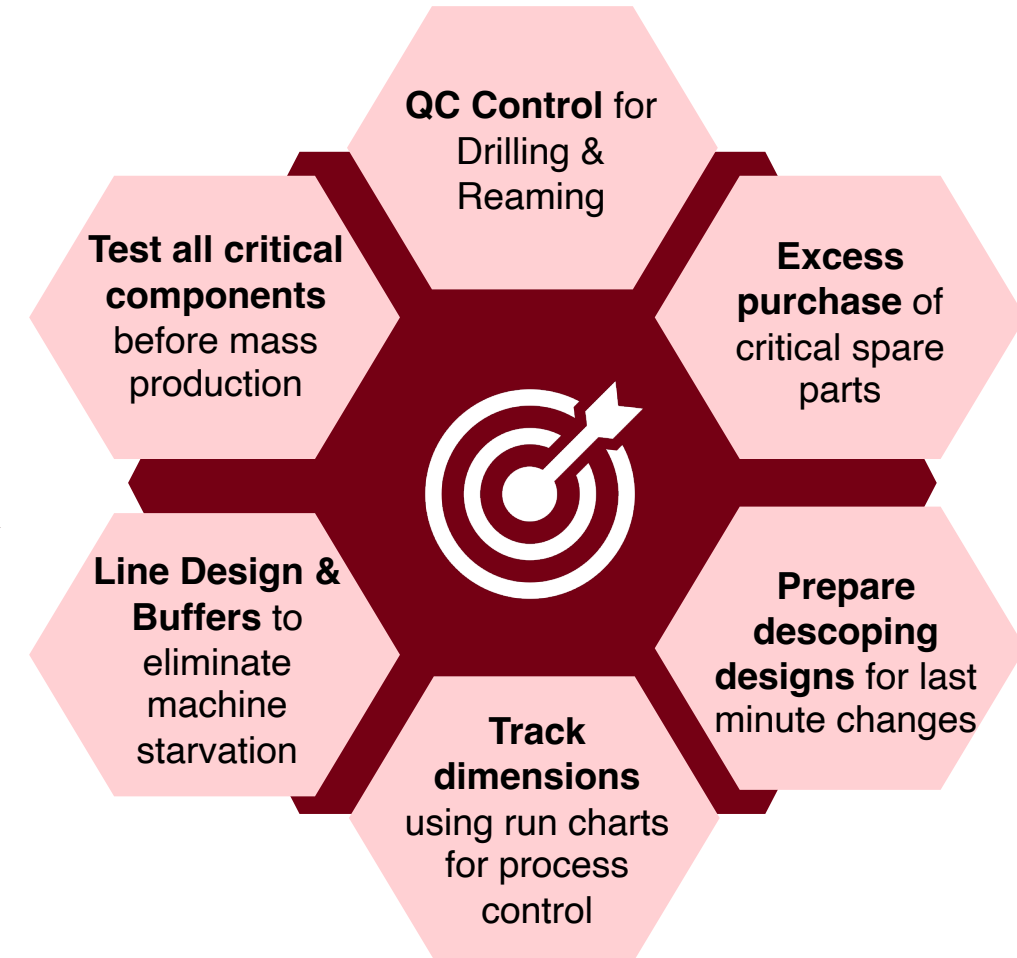
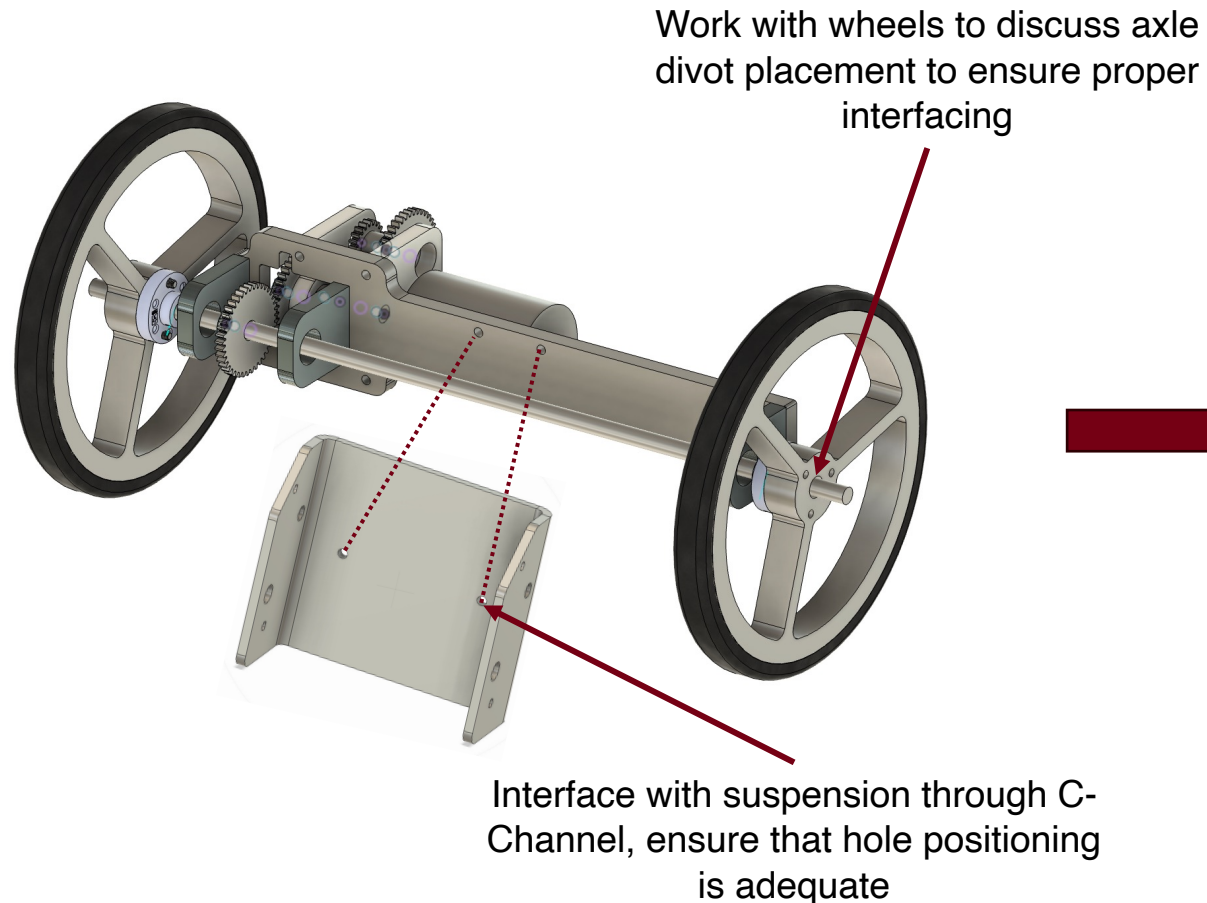
Descoping Strategies

Risk	Impact	Descope Strategy
Computer Vision System not achieving repeatable accuracy	Delays quality verification for batch production	Switch to manual measurement checklist / ream all axle holders
Breaking of Taps	Delays production waiting on tooling	Buy and stock extra taps, redesign parts without the use of taps

Critical Features and Strategic Plan

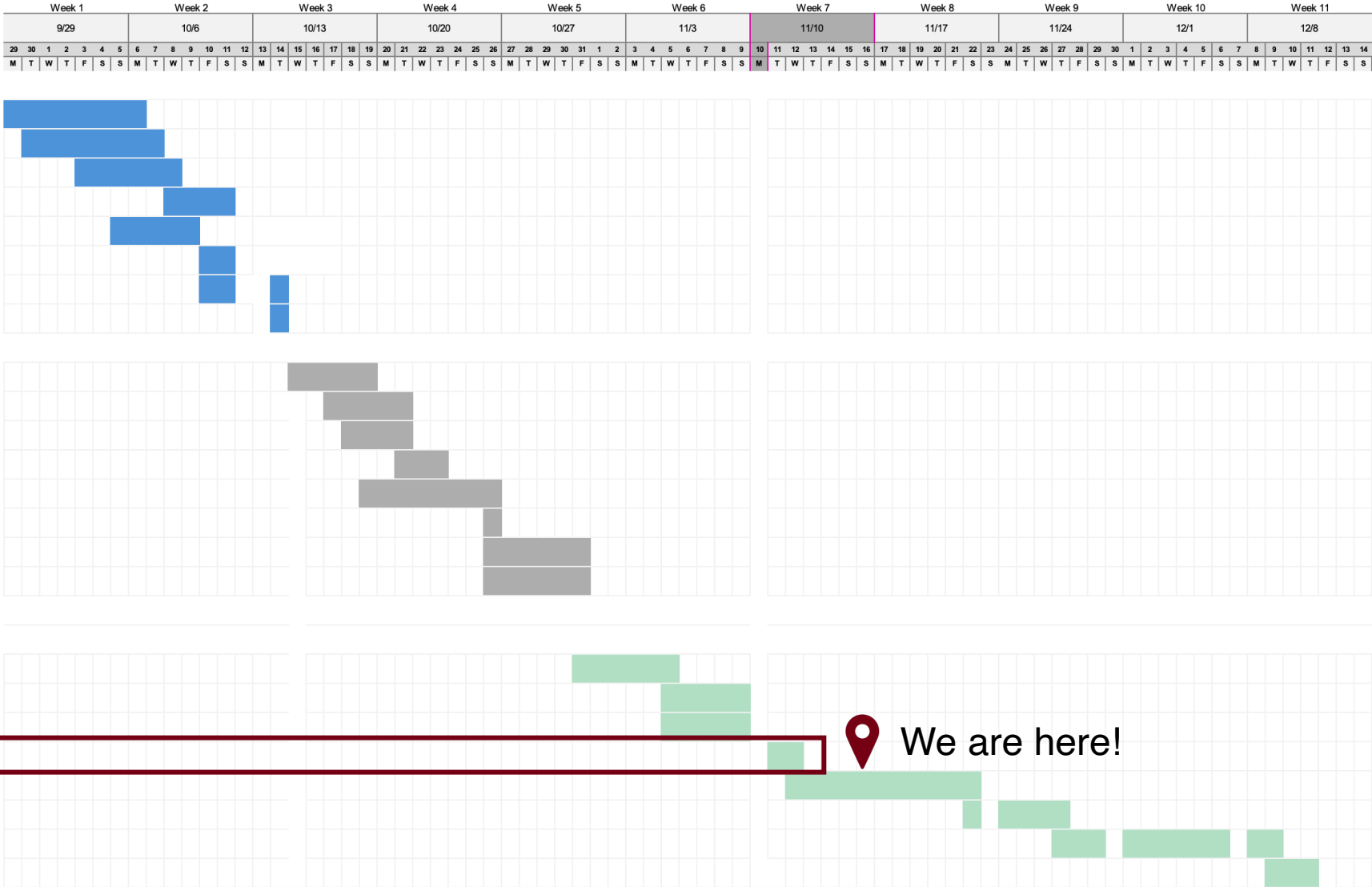
Efficiently fabricating 40 RC Cars while ensuring high assembly quality

Drivetrain primarily interfaces with wheels and suspension



Gantt Chart

TASK	START	END
Phase 1: Design & Analysis		
Sketch Drivetrain Design	9/29	10/6
Identify Constraints (Shaft Size, Gear Options, Torque)	9/30	10/7
Preliminary CAD Models	10/3	10/8
Perform Torque and Gear Ratio Analysis	10/8	10/11
Preliminary Materials List (BOM) & Budget	10/5	10/9
Create Manufacturing CAD Model (Adjusting for Extra Tolerances)	10/10	10/12
Finalized Detailed CAD and Dimensioned Drawings	10/10	10/15
Design Review Presentation	10/15	10/15
Prototype Fabrication & Validation		
3D Print finalized DriveTrain subassembly	10/15	10/19
Cut components via Waterjet	10/17	10/21
Drill, Ream, and Tap Critical Holes	10/18	10/21
Test Fit & Functional Prototype Validation	10/21	10/23
Adjust Design Tolerances & Interfacing Issues in CAD	10/19	10/26
Working Prototype of Sub-Assembly	10/26	10/26
All Teams Prototype Car Assembly	10/26	10/31
Developed Preliminary Manufacturing Operation Sheets	10/26	10/31
Phase 3: Batch Manufacturing, Assembly, & Testing		
Design and Machine Assembly Jigs and/or Fixture	10/31	11/2
Finalized Manufacturing Drawings & Operation Sheets	10/31	11/5
Waterjet Cutting of Early Manufacturing Batch	11/5	11/10
Sanding, Machining, Drill/Tap/Ream Axle Holes for Preliminary Batch	11/5	11/10
Manufacturing Review Presentation	11/10	11/12
Finalized Batch Manufacturing	11/12	11/22
Assembly & Integration	11/22	11/27
Fine-Tuning & Full-Vehicle Testing	11/27	12/9
RC Car Race Event	12/9	12/11



Questions?

Appendix A: Low Level Process Plan

Axle Holder

#	Task	Machine	Tooling	Cycle Time (min:sec)
00	Cut stock to shape	Water jet	Water jet	0:29
01	Check the quality of the bearing hole	RasPi Camera	RasPi Camera	0:02
02	Ream Bearing Holes If < 0.498 in.	Mill 1	0.498 in. reamer	1:42
03	Face the bottom of the workpiece	Mill 2	½ in. endmill	1:56
04	Center drill holes	Mill 3	#2 60 degrees Center drill	0:38
05	Drill holes	Mill 4	#43 drill	2:26
06	Tap holes	Powered tap	4-40 tap	1:03

Appendix A: Low Level Process Plan

Gear Block

#	Task & Questions	Machine	Tooling	Cycle Time (min:sec)
00	Cut stock to shape	Water jet	Water jet	1:01
01	Check the quality of the bearing hole	RasPi Camera	RasPi Camera	0:02
02	Ream Bearing Holes If < 0.498 in.	Mill 1	0.498 in. reamer	2:17
03	Face the bottom of the workpiece	Mill 2	½ in. endmill	1:50
04	Center drill holes	Mill 3	#2 60 degrees Center drill	0:19
05	Drill holes	Mill 4	#43 drill	2:27
06	Tap holes	Powered tap	4-40 tap	1:06

Appendix A: Low Level Process Plan

Motor Mount

#	Task & Questions	Machine	Tooling	Cycle Time (min:sec)
00	Cut stock to shape	Water jet	Water jet	0:57
01	Check the quality of the bearing hole	RasPi Camera	RasPi Camera	0:02
02	Ream Bearing Holes If < 0.498 in.	Mill 1	0.498 in. reamer	2:06
03	Face the bottom of the workpiece	Mill 2	½ in. endmill	2:20
04	Center drill holes	Mill 3	#2 60 degrees Center drill	0:28
05	Drill holes	Mill 4	#43 drill	2:23
06	Tap holes	Powered tap	4-40 tap	1:05

Appendix A: Low Level Process Plan

Base

#	Task & Questions	Machine	Tooling	Cycle Time (min:sec)
00	Cut stock to shape	Water jet	Water jet	3:47
01	Remove nub	Belt sander	Belt sander	0:59

Appendix A: Low Level Process Plan

Gears

#	Task & Questions	Machine	Tooling	Cycle Time (min:sec)
00	Ream out gear	Drill press	1/4 in. reamer Gear reaming jig	0:54

Short + Medium Axles

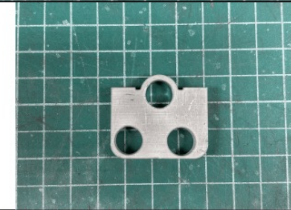
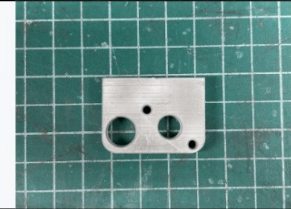
#	Task & Questions	Machine	Tooling	Cycle Time (min:sec)
00	Cut the axle to length	Vertical band saw	Vertical band saw blade Axle cutting jig	0:51
01	Sand axle ends	Belt sander	Belt sander	1:03

Appendix A: Low Level Process Plan

Drive Axle

#	Task & Questions	Machine	Tooling	Cycle Time (min:sec)
00	Fixture and locate axle stock in mill	Mill	Collet holder, soft jaws, edge finder,	1:29
01	Center drill gear and wheel divet	Mill	#2 60 degrees Center drill	0:42
02	Refixture the axle stock with the machined end in the collet.	Mill	Collet holder, soft jaws	1:06
03	Center drill wheel divet	Mill	#2 60 degrees Center drill	0:29

Appendix B: Bill of Materials (BOM)

Bill of Materials							
Part Number	Part Image	Description	Quantity/Area (in^2)	Material	Cost	Total Cost	Source/Notes
Base		The base is the central component that all other drivetrain components connect to. The base is also the means by which the suspension connects to the drivetrain.	9.73	1/4 in. 6061 Aluminum	\$ 0.52	\$ 5.06	Provided by 2.810 or https://www.mcmaster.com/89155K11/
Axle Holder		Supports the main axle. Mounts to the base.	2.18	1/4 in. 6061 Aluminum	\$ 0.52	\$ 1.13	Provided by 2.810 or https://www.mcmaster.com/89155K11/
Gear Block		Supports the bearings and axles that hold the 2nd and 3rd gear reductions. Mounts to the base.	2.27	1/4 in. 6061 Aluminum	\$ 0.52	\$ 1.18	Provided by 2.810 or https://www.mcmaster.com/89155K11/
Motor Mount		Supports the bearings and axles that hold the 1st gear reduction. The motor also mounts to this component. This part mounts to the base.	1.23	1/4 in. 6061 Aluminum	\$ 0.52	\$ 0.64	Provided by 2.810 or https://www.mcmaster.com/89155K11/

Appendix B: Bill of Materials (BOM)

Bill of Materials								
Part Number	Part Image	Description	Quantity/Area (in^2)	Material	Cost	Total Cost	Source/Notes	
12 Tooth Gear		The smallest gear used in the drivetrain. This gear attaches to the motor shaft.	1	Stainless Steel Gear Stock	\$ -	\$ -	Provided by 2.810	
20 Tooth Gear		The medium sized gear used in the drivetrain. This gear is used in the 2nd and 3rd gear reductions.	2	Stainless Steel Gear Stock	\$ -	\$ -	Provided by 2.810	
40 Tooth Gear		The largest gear used in the drivetrain. This gear is used in all gear reduction stages as well as the final drive.	4	Stainless Steel Gear Stock	\$ -	\$ -	Provided by 2.810	
Short axle (26.4mm)		Used to connect the the 2nd and 3rd gear reduction stages and the final drive.	2	Stainless Steel D-Shaft	\$ -	\$ -	Provided by 2.810	

Appendix B: Bill of Materials (BOM)

Bill of Materials								
Part Number	Part Image	Description	Quantity/Area (in^2)	Material	Cost	Total Cost	Source/Notes	
Medium axle (33.25mm)		Used to connect the first and second gear reduction stages.	1	Stainless Steel D-Shaft	\$ -	\$ -	Provided by 2.810	
Drive Axle (304mm)		Used as the main drive axle	1	Stainless Steel D-Shaft	\$ -	\$ -	Provided by 2.810	
Bearings		Used to ensure smooth rotation of all axles.	8	Chrome Steel	\$ 0.80	\$ 6.39	https://tinyurl.com/42vrtkr3 *Lead Time: ~ 1-2 days	
Shaft Collar		Used to prevent axles from moving side to side.	3	Zinc-Plated 1215 Carbon Steel	\$ 1.83	\$ 5.49	https://www.mcmaster.com/6432K12/ *Lead Time: ~ 2 days	

Appendix B: Bill of Materials (BOM)

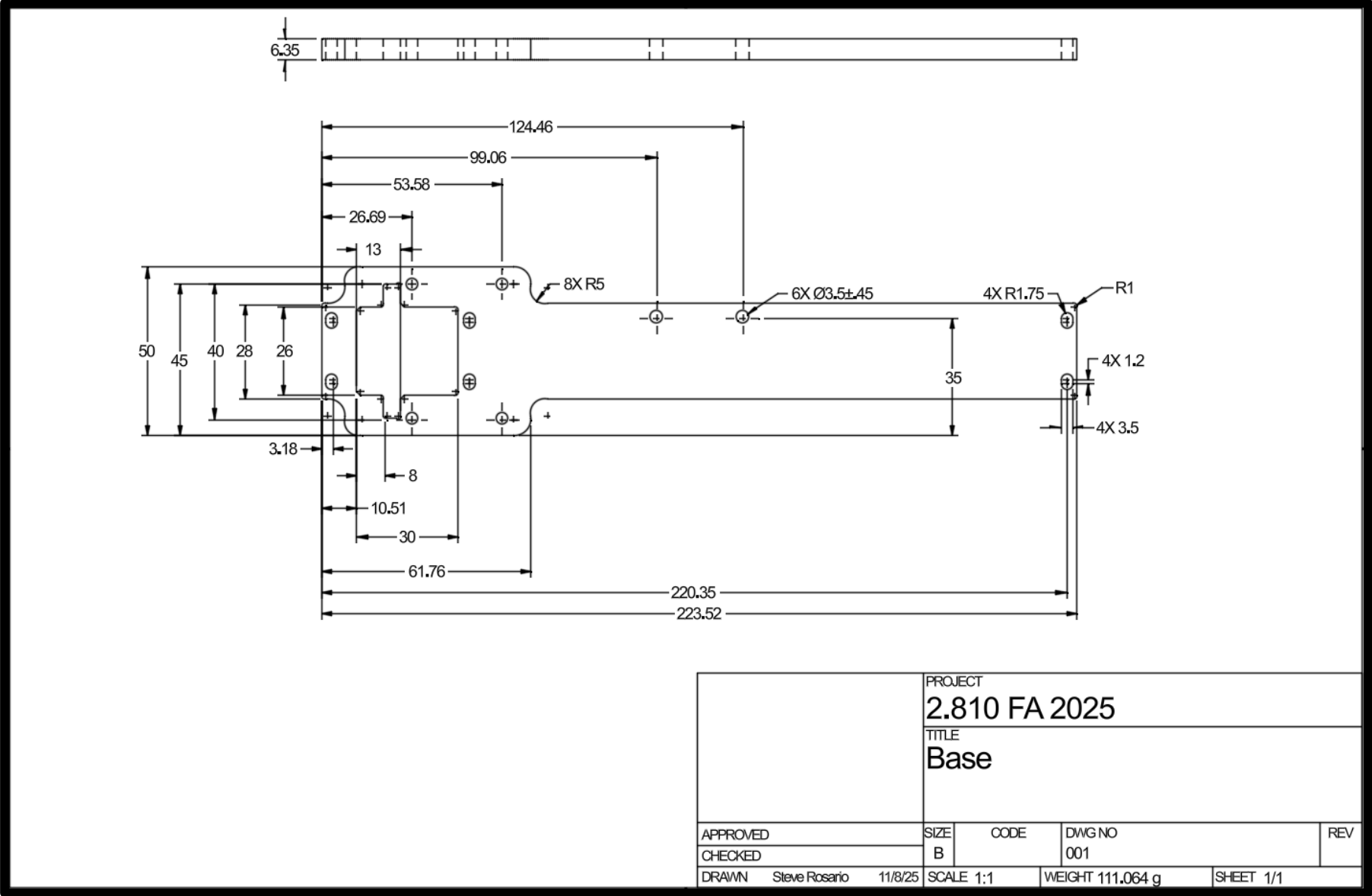
Bill of Materials							
Part Number	Part Image	Description	Quantity/Area (in^2)	Material	Cost	Total Cost	Source/Notes
10-32 Set Screw		Used to affix the 20 tooth gears, 40 tooth gears, and shaft collars to the axles.	8	18-8 Stainless Steel	\$ 0.08	\$ 0.67	https://www.mcmaster.com/92311A422/ *Lead Time: ~ 2 days
4-40 x 1/8 Set Screw		Used to affix the 12 tooth gear to the motor shaft.	1	18-8 Stainless Steel	\$ 0.09	\$ 0.09	https://www.mcmaster.com/94355A130/ *Lead Time: ~ 2 days
4-40 x 1/8 Cone Point Set Screw		Used to secure the drive gear to the drive axle	1	18-8 Stainless Steel	\$ 0.25	\$ 0.25	https://www.mcmaster.com/92785A523/ *Lead Time: ~ 2 days
4-40 x 1/2 Screw		Used to connect the axle holders, gear block, motor mount, and motor to the base.	14	18-8 Stainless Steel	\$ 0.06	\$ 0.86	https://www.mcmaster.com/92196A110/ *Lead Time: ~ 2 days

Appendix B: Bill of Materials (BOM)

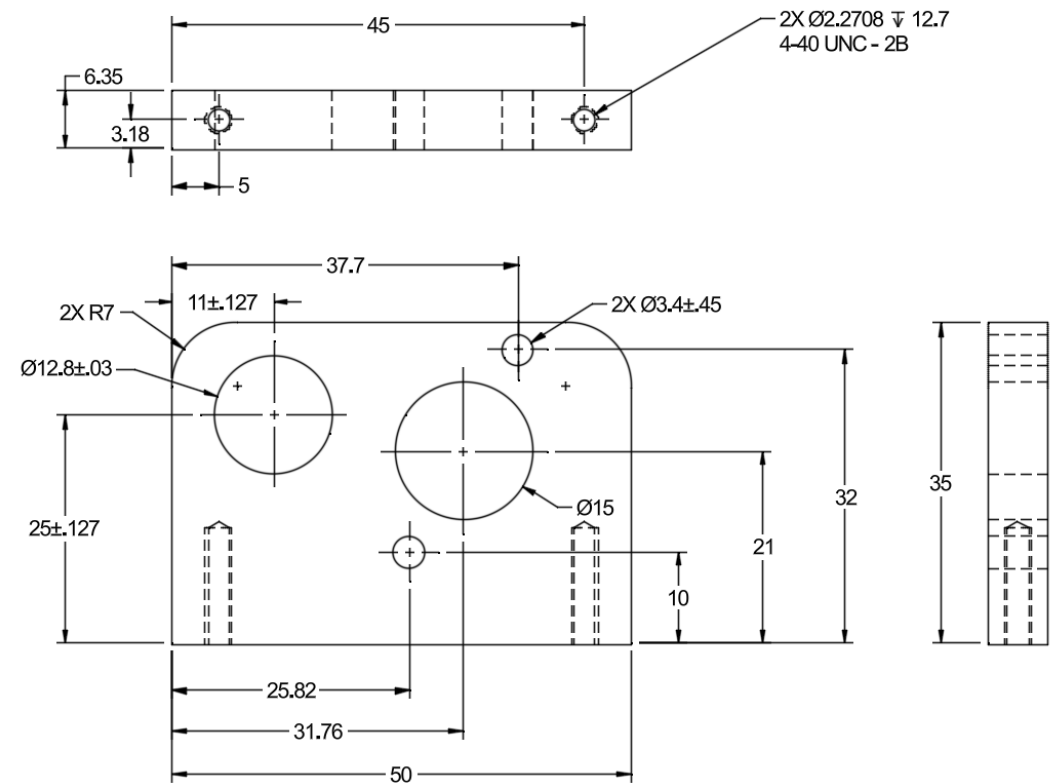
Bill of Materials							
Part Number	Part Image	Description	Quantity/Area (in^2)	Material	Cost	Total Cost	Source/Notes
Bullet connectors		Used to connect the motor wires to the motor controller	4		\$ 0.09	\$ 0.36	https://www.amazon.com/Feggizuli-Connectors-Insulation-Aut *Lead Time: ~ 2 days
16 AWG Wire		Connects the motor wires with wires from the motor controller	8		\$ 0.02	\$ 0.17	https://www.amazon.com/Silicone-Flexible-Temperature-Resis *Lead Time: ~ 2 days
DC Brushed Motor		Used to power the RC car.	1		\$ -	\$ -	Provided by 2.810 https://tinyurl.com/krkwujda
6061 1/4in Sheet Stock		Used to cut out metal parts	6		\$ -	\$ -	Provided by 2.810 https://www.mcmaster.com/89155K11-891 *Lead Time: ~ 2 days
				Total Cost	\$ 22.31		

Bill of Materials

Appendix C: Drawings

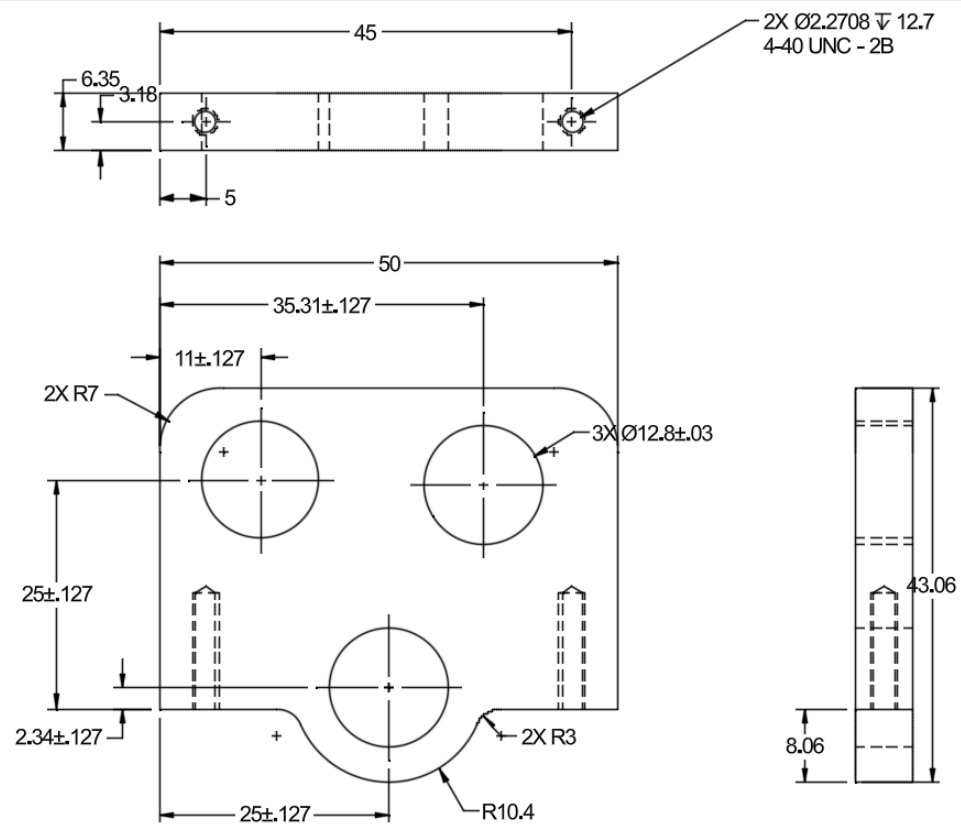


Appendix C: Drawings



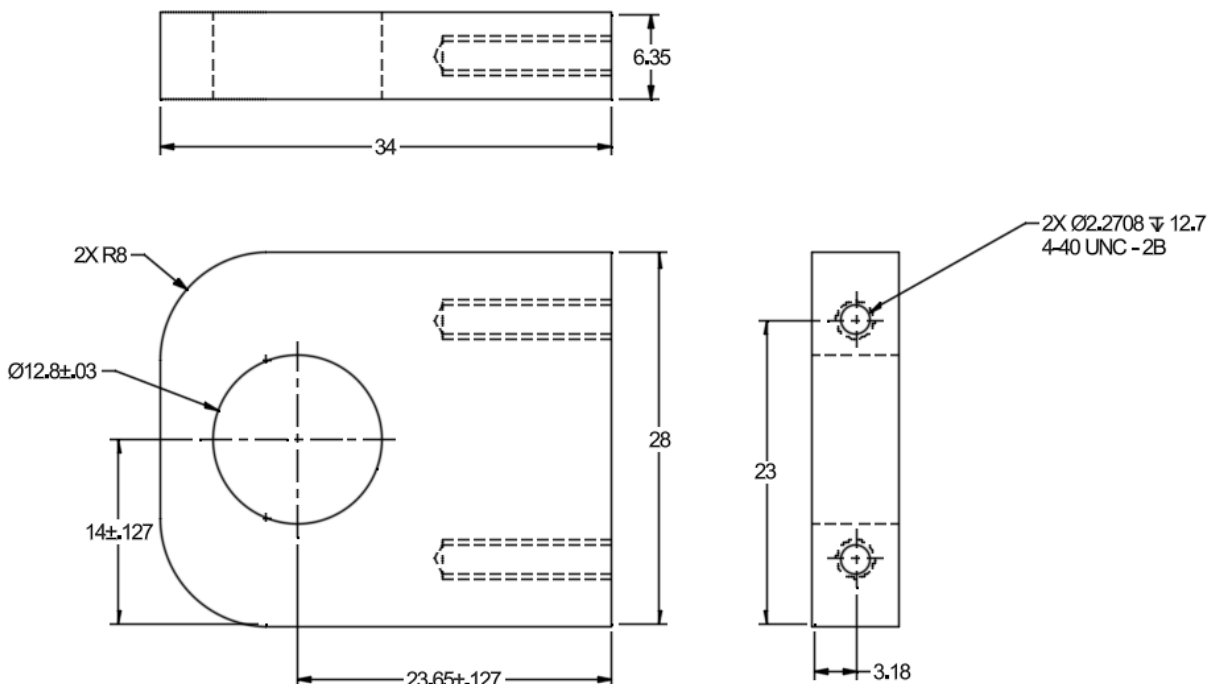
		PROJECT			
		2.810 FA 2025			
		TITLE			
		Motor Mount			
APPROVED	SIZE	CODE	DWG NO	REV	
CHECKED	B		002		
DRAWN	Steve Rosario	11/8/25	SCALE 2:1	WEIGHT 23.813 g	SHEET 1/1

Appendix C: Drawings



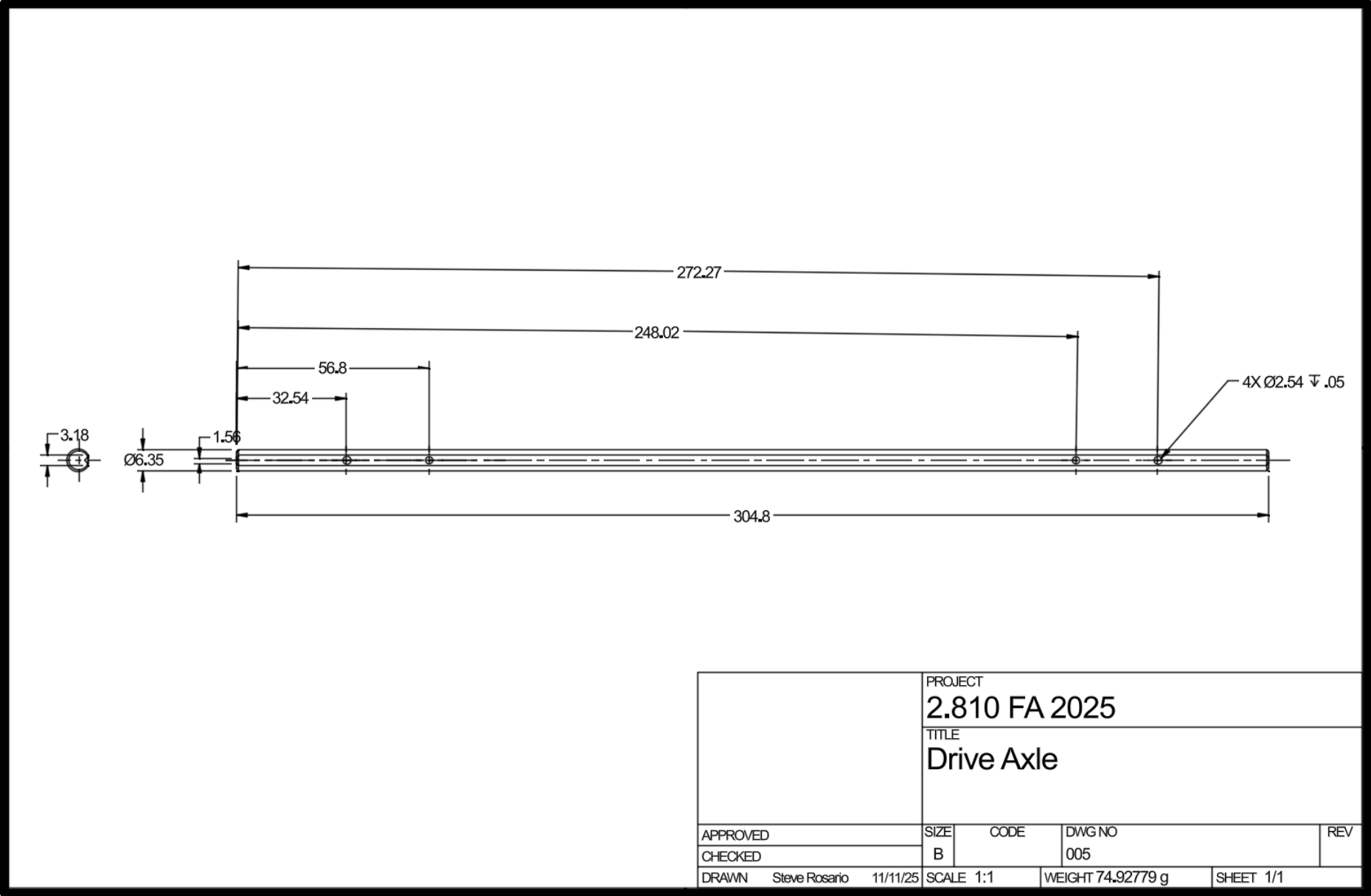
		PROJECT		
		2.810 FA 2025		
		TITLE		
		Gear Block		
APPROVED	SIZE	CODE	DWG NO	REV
CHECKED	B		003	
DRAWN	Steve Rosario	11/8/25	SCALE 2:1	WEIGHT 24.855 g
			SHEET 1/1	

Appendix C: Drawings

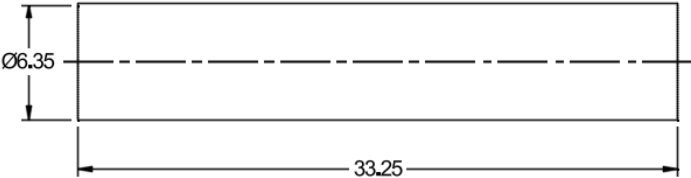


		PROJECT 2.810 FA 2025			
		TITLE Axle Holder			
APPROVED	SIZE	CODE	DWG NO	REV	
CHECKED	B		004		
DRAWN	Steve Rosario	11/8/25	SCALE 3:1	WEIGHT 13.362 g	SHEET 1/1

Appendix C: Drawings



Appendix C: Drawings



		PROJECT		
		2.810 FA 2025		
		TITLE		
		Medium Axle		
APPROVED	SIZE	CODE	DWG NO	REV
CHECKED	B		006	
DRAWN	Steve Rosario	11/11/25	SCALE 4:1	WEIGHT 8,424 g
		SHEET 1/1		

Appendix C: Drawings



		PROJECT			
		2.810 FA 2025			
		TITLE			
		Small Axle			
APPROVED		SIZE	CODE	DWG NO	REV
CHECKED		B		007	
DRAWN	Steve Rosario	11/11/25	SCALE 8:1	WEIGHT 6.689 g	SHEET 1/1

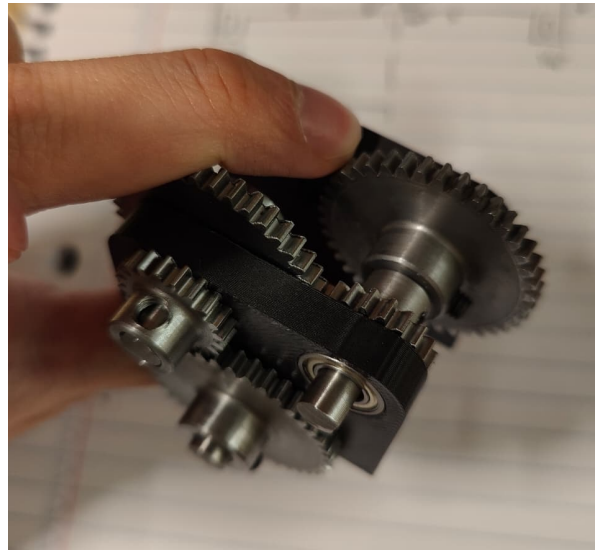
Appendix D: Design Review References

Gear Ratio Analysis & Design Decisions

Objective: Based on motor specifications, achieve an adequate torque multiplication for navigating obstacles and ramp while maintaining a reasonable wheel speed.

Motor Constraints

- **Motor:** Reddy Radon 2, 17T, 3600Kv
- **Input Voltage:** 7.4 V LiPo
- **Motor RPM:** 26,600 RPM (3600 x 7.4V)
- Motor shaft diameter limits it to a 12T pinion gear



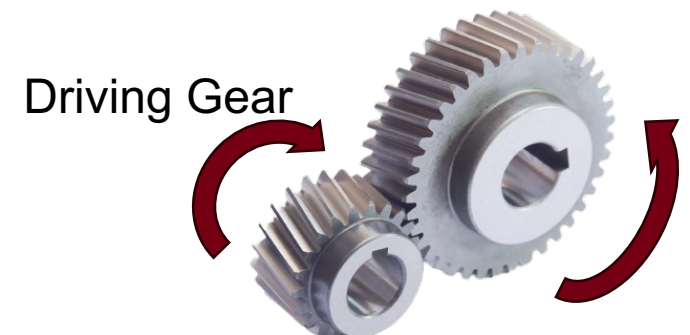
Note: RPM & Speed Calculations assume theoretical maximum (friction, car mass, gravity will slow the speed down by ~60-70%)

Gear Reduction Design

- **13.33x Reduction:** 12 → 40, 20 → 40, 20 → 40
- **Wheel RPM (no-load):** ~1998 RPM
- **Motor RPM:** 26,600 RPM (3600 x 7.4)
- **No-Load Linear Speed (Assume D = 6 in):** 15.95 m/s

↓ Speed $\text{Gear Reduction} = \frac{\text{Driven}}{\text{Driving}}$ ↑ Torque

Driven Gear



Appendix D: Design Review References

Ramp Climbing Feasibility Analysis

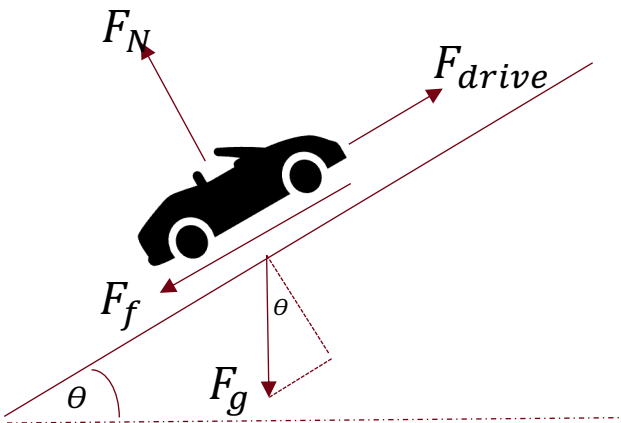
Objective: Determine whether the motor provides sufficient torque for the vehicle to climb the ramp based on certain assumptions.

$$F_{drive} = mgsin\theta + \mu mgcos\theta$$

$$T_{axle,req} = \frac{F_{drive}r_{wheel}}{Gear\ Reduction}$$

$$T_{motor} = \frac{P}{\omega} = \frac{21.6W}{(26640\ RPM * \frac{2\pi}{60})} = 0.00774\ Nm$$

$$T_{axle,available} = T_{motor} * Gear\ Reduction$$



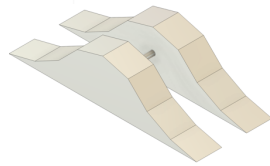
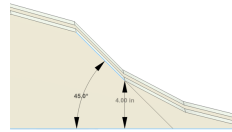
Assumptions

- Wheel diameter (D = 6")
- Gravity (g = 9.81 m/s²)
- RC car mass cases: 3 lbs., 5 lbs.
- Friction coefficient: 0.1
- Maximum Ramp Angle: 45°
- Max Battery Power: 21.6 W

Mass	$F_{gx} + F_f$ (N)	$T_{axle,req}$ (Nm)	$T_{axle,avail}$ (Nm)
3 lbs. (~1.36 kg)	10.35	0.059	0.103
5 lbs. (~2.27 kg)	17.28	0.099	0.103

Appendix D: Key Design Constraints

Engineering Constraints → Design Requirements

	Engineering Constraints	Design Requirements
Motor Shaft Diameter 	Motor Shaft fits only 12T gear	Must use 12-tooth pinion on motor
Axle Diameter 0.25" 	Spur gears fit 0.25" axle only in 20T/40T	Limited compatible gear combos, only 12:40 or 20:40
Ramp Width 	Ramp width 5.25" to 16"	Axle length must fit within 5.25" to 16"
Maximum Ramp Angle 	Max Rap Angle 45 degrees	Provide sufficient torque to climb slope
Vehicle Weight 	Vehicle Mass ~3-5 lbs.	Drivetrain must support load and ensure traction